JAYA PRAKASH NARAYAN COLLEGE OF ENGINEERING

DHARMAPUR, MAHABUBNAGAR – 509001

UGC Autonomous Institution, Accredited By NAAC, Approved By AICTE and Affilitated To JNTUH , Telangana , India

"360° Blind Spot"

A Dissertation on Real Time Research Project submitted to the Jawaharlal Nehru Technological University, Hyderabad in partial fulfillment of the requirement for the award of degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

Submitted By

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2025-2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that Real Time Research Project report on **"360° Blind Spot"** is a bonafide work done by **C. Kushi (23361A0535).**In partial fulfillment of the requirement of the award for the degree of Bachelor of Technology in **"Computer Science and Engineering" JNTUH,** Hyderabad during the year 2025-2026.

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III. ACKNOWLEDGEMENT

We owe a debt of gratitude to **Dr. K. Guru Raghavendra Reddy** , CSE, JPNCE for his admirable guidance an inspirational both theoretically and practically and most importantly for the drive to complete projects successfully. Working under such an eminent guide was our privilege.

We express our sincere thanks to **Mrs. K.RADHIKA**, **Incharge,** CSE, JPNCE of all kinds of consideration, support and encouragement in carrying out this project successfully.

We would also like to thank **Dr. PANNALA KRISHNA MURTHY**, **Principal**, JPNCE for his cooperation and encouragement.

We are grateful to the department of Computer Science and Engineering for providing us with excellent lab and library facilities.

We thank our parents for the love, care and moral support without which we would have not been able to complete this project. It has been a constant source of inspiration for all my academic endeavor.

PROJECT ASSOCIATE

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IV. ABSTRACT

Visually impaired individuals often face safety risks due to obstacles around them. Most existing systems detect obstacles only in limited directions, creating blind spots and increasing accident risk. They also lack instant alerts like vibration or buzzer warnings.

The proposed 360° Blind Spot system uses ultrasonic sensors to detect obstacles in all directions. A micro-controller processes the data in real time and alerts the user through a buzzer when an obstacle is detected.

The proposed system serves as an assistive navigation aid for visually impaired users and a safety tool for elderly people. It detects obstacles during indoor and outdoor movement, improving mobility, confidence, and overall safety, especially in crowded or unfamiliar environments.

It provides complete obstacle detection with instant alerts. Using continuous scanning, buzzer, it improves user safety and supports independent mobility.

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1. INTRODUCTION

1.1 Background

Visually impaired individuals often face difficulties in detecting obstacles around them. Most existing systems can detect objects only in one or two directions, leaving blind spots. These limitations increase the risk of accidents. Hence, a system that can monitor all directions and give instant alerts is needed.

1.2 Purpose

The purpose of this system is to detect obstacles in all directions and alert the user using a buzzer sound. It aims to improve safety and help users avoid collisions while walking.

1.3 Scope

This system can be used as:

- An assistive device for visually impaired people
- A safety tool for elderly individuals
- An obstacle detection system for indoor and outdoor environments

1.4 Methodology

- Ultrasonic sensors are placed on the device to cover the directions.
- These sensors continuously detect the distance of nearby objects.
- A microcontroller processes the sensor data.
- When an obstacle is detected within a certain range, the system activates a buzzer.
- The buzzer alerts the user immediately, helping them avoid the obstacle.

2. SYSTEM ANALYSIS

2.1 Existing System

- Uses ultrasonic, radar, or camera sensors
- Detects obstacles only in limited directions

2.1.1 Disadvantages

- Not full coverage (has blind spots)
- Expensive (radar/camera systems)
- Affected by weather and lighting
- May give wrong or delayed alerts

2.2 **Proposed System**

- Uses ultrasonic sensors
- Covers full area
- Gives alert using a buzzer sound

2.2.1 **Advantages**

- Full detection (no blind spots)
- Low cost and simple design
- Quick buzzer alert
- Easy to use and portable

3. SYSTEM REQUIREMENTS

3.1 Hardware requirements

3.1.1 ESP32 Development Board:

The ESP32 acts as the main controller of the system and controls all operations. It receives signals from the ultrasonic sensors and processes the data to detect obstacles. Based on this information, it activates the buzzer to alert the user. It ensures fast and accurate response, making the system reliable. Its compact size and efficiency make it suitable for wearable projects. Overall, it works as the brain of the entire system.



Fig 3.1 -ESP32

3.1.2 HC-SR04 Ultrasonic Sensors:

The ultrasonic sensors are used to detect obstacles around the user. They send sound waves and measure the distance of nearby objects. By placing multiple sensors in different directions, the system achieves full coverage. These sensors continuously scan the surroundings for safety. They are low-cost and easy to use, making them ideal for this project. Their role is very important for accurate obstacle detection.



Fig 3.2 -HC-SR04 Ultrasonic sensor

3.1.3 Active Buzzer Module:

The buzzer is the main alert component in the system. It produces a sound whenever an obstacle is detected within a certain distance. This helps the user react quickly and avoid collisions. It is simple to use and does not require complex programming. The sound alert is clear and immediate. It plays a key role in ensuring user safety.



Fig 3.3 - Active Buzzer Module

3.1.4 Glasses:

The components are mounted on a glasses to make the system wearable and hands-free. The ultrasonic sensors are placed in different directions one at the front, and the others on the left and right sides so they can detect obstacles all around the user. The ESP32 board, battery, and resistors are usually fixed on the top or back side of the glasses to keep the weight balanced and comfortable. The buzzer is placed near the ear so the user can clearly hear the alert sound. If a vibration motor is used, it can be placed on the side of the frame to provide quick physical feedback. All components are connected using jumper wires and fixed securely to avoid movement. This arrangement ensures proper detection while keeping the device easy to wear and use.

3.1.5 Jumper Wires & PCB/Breadboard:

Jumper wires are used to connect all the components together. A breadboard is used for testing the circuit, while a PCB is used for permanent setup. These components help in building and organizing the system neatly. They ensure proper electrical connections between parts. Without them, the system cannot function. They are essential for assembling the project.

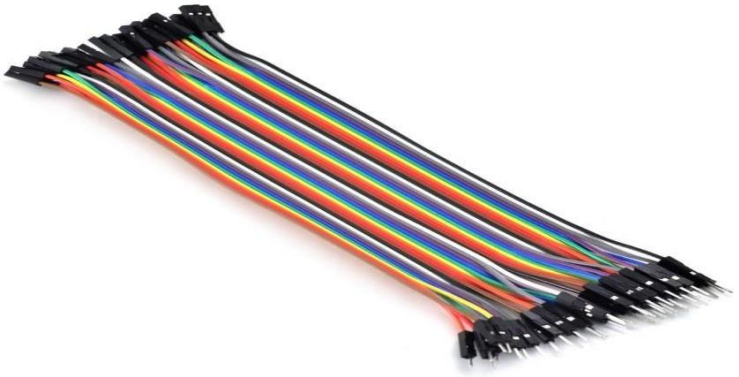


Fig 3.6 -Jumper wires

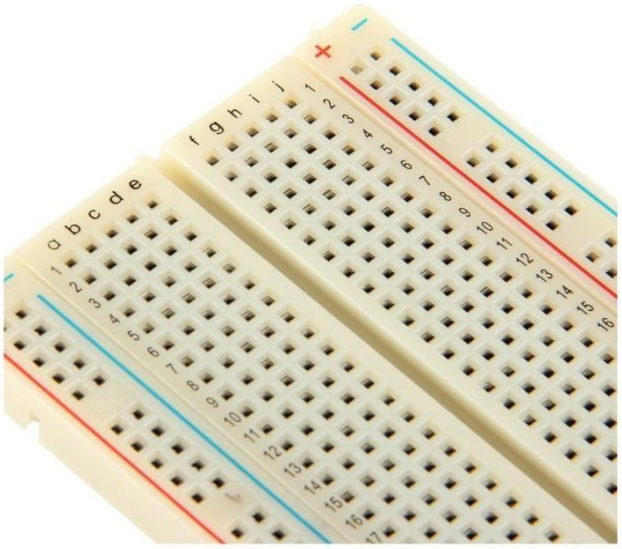


Fig 3.7 -Breadboard

3.1.6 IR Obstacle Detection Sensor Module:

IR obstacle detection sensor module, which is used to detect objects that are very close to the device. It works based on the principle of infrared light reflection, where an IR LED emits invisible infrared rays and a photodiode receives the reflected light when it hits an obstacle. When an object comes within a certain distance, the sensor detects the reflected signal and sends a digital output to the controller, such as the ESP32 microcontroller. This helps your system quickly identify nearby obstacles that ultrasonic sensors might miss, especially at very short ranges, and triggers alerts like a buzzer or vibration to warn the user, improving overall safety and responsiveness.

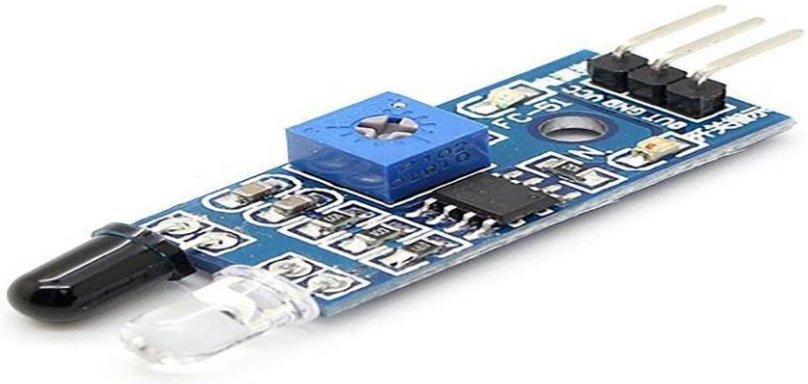


Fig 3.9 -IR obstacle detection sensor module

3.2 Software requirements

3.2.1 Arduino IDE:

The Arduino IDE is used to write and upload the program to the ESP32 board. In this project, the code is written to control the ultrasonic sensors, calculate distance, and activate the buzzer when an obstacle is detected. It provides a simple interface for coding, compiling, and uploading the program through a USB cable. Libraries required for the sensors can also be added easily. It is mainly used for testing and running the hardware.

Pin Configuration:

Left sensor -trig:5 | echo: 18
Right sensor -trig: TX2 | echo: RX2
Ir sensor - out | D13
Buzzer: 19

Complete Arduino Code:

```
#define TRIG_FRONT 5
#define ECHO_FRONT 18
#define TRIG_SIDE 17
#define ECHO_SIDE 16
#define IR_SENSOR 4
#define BUZZER 19
long readUltrasonic(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH, 30000); // timeout 30ms
  long distance = duration * 0.034 / 2;
  return distance;
}
void setup() {
  Serial.begin(115200);
  pinMode(TRIG_FRONT, OUTPUT);
  pinMode(ECHO_FRONT, INPUT);
  pinMode(TRIG_SIDE, OUTPUT);
  pinMode(ECHO_SIDE, INPUT);
  pinMode(IR_SENSOR, INPUT);
```

```
  pinMode(BUZZER, OUTPUT);
}
void loop() {
  long front = readUltrasonic(TRIG_FRONT, ECHO_FRONT);
  long side  = readUltrasonic(TRIG_SIDE, ECHO_SIDE);
  int ir = digitalRead(IR_SENSOR);
  // NOTE: IR = 0 → obstacle detected
  //       IR = 1 → no obstacle
  //        Print values (for testing)
  Serial.print("Front:  ");  Serial.print(front);
  Serial.print(" cm  | Side: "); Serial.print(side);
  Serial.print(" cm  | IR: "); Serial.println(ir);
   if ((front > 0 && front < 50) ||
      (side > 0 && side < 50) ||
      (ir == 0)) {
    digitalWrite(BUZZER, HIGH);   // ALERT ON
  }
  else {
    digitalWrite(BUZZER, LOW);    // ALERT OFF
  }
  delay(200);
}
```

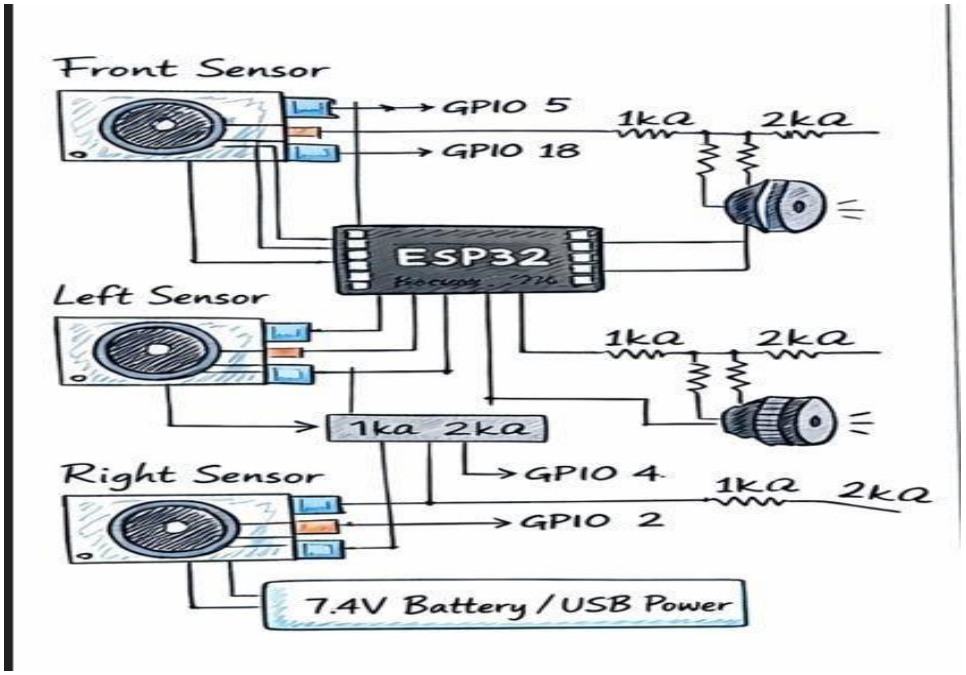
3.2.2 VS code:

VS Code is used as an advanced coding environment for writing and managing the project code. With extensions like PlatformIO or Arduino, it provides better features such as auto-completion, error checking, and organized project files. It helps in writing cleaner and more efficient code for the ESP32. VS Code is useful for debugging and improving the program. It makes development easier, especially for larger or more complex projects.

4. SYSTEM DESIGN

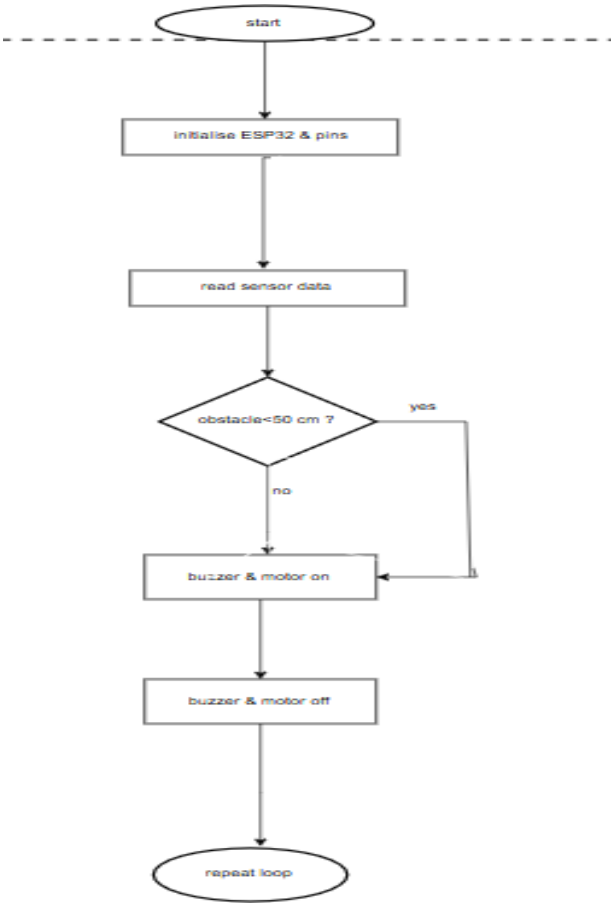
4.1 Circuit diagram:

The circuit diagram shows how all components are connected to the esp32. Three ultrasonic sensors are placed at the front, left, and right, and their trig and echo pins are connected to gpio pins (5,18), (17,16), and (4,2). The echo pins use resistors to reduce voltage and protect the esp32. The buzzer (pin 19) and vibration motor (pin 23) are connected as output devices. A battery or power bank provides power to the system. This setup ensures proper connection and functioning of all components.



4.2 Flow diagram

The flow diagram explains the step-by-step working of the system. First, the system starts and initializes the ESP32 and pins. Then, it reads the distance from all sensors. If any obstacle is detected within 50 cm, the buzzer and vibration motor are turned ON. If no obstacle is detected, they remain OFF. This process keeps repeating continuously for real-time detection and safety.



5. IMPLEMENTATION

The implementation of the 360° blind spot detection system involves assembling the hardware, programming the ESP32, and testing the device. First, all components such as the ESP32, ultrasonic sensors, buzzer, vibration motor, and resistors are connected according to the circuit diagram. The sensors are positioned at the front, left, and right to ensure full coverage.

Next, the program is written in Arduino IDE or VS Code to read distance values from the sensors and control the buzzer and vibration motor based on the set threshold (50 cm). The code is then uploaded to the ESP32 board.

After programming, the system is tested to check whether obstacles are detected correctly and alerts are given properly. Finally, all components are mounted on a glasses structure to make the device wearable, and real-time testing is done to ensure proper working and user safety.

6. RESULT

The 360° blind spot detection system was successfully developed and tested. The ultrasonic sensors were able to detect obstacles in the front, left, and right directions effectively. Whenever an object was detected within 50 cm, the buzzer and vibration motor were activated immediately, providing a quick alert to the user.

The system showed reliable performance in both indoor and outdoor environments. It provided continuous monitoring and helped in avoiding collisions. The device was lightweight, portable, and easy to use when mounted on a headband or glasses.

Overall, the project achieved its goal of providing a simple, low-cost, and efficient solution for obstacle detection and user safety.

7. CONCLUSION & FUTURE SCOPE

7.1 Conclusion

The 360° blind spot detection system was successfully designed and implemented using ultrasonic sensors and an ESP32. The system effectively detects obstacles in directions and provides immediate alerts through a buzzer and vibration motor. It is simple, low-cost, and easy to use, making it suitable for visually impaired individuals and general safety applications. Overall, the project improves user awareness and helps in avoiding accidents, achieving its main objective.

7.2 Future scope

In the future, the system can be improved by adding voice alerts instead of only buzzer sound for better understanding. More sensors can be added to achieve complete 360° coverage without any gaps. Integration with a mobile application using Bluetooth or Wi-Fi can provide additional features like notifications and tracking. The system can also be enhanced using AI-based object detection for better accuracy. Furthermore, reducing size and improving battery life can make the device more comfortable and efficient for daily use.

8. REFERNCES

1. Books on Embedded Systems and IoT for understanding ESP32 and sensors
2. Arduino Official Documentation – https://www.arduino.cc
3. ESP32 Documentation by Express if – https://www.espressif.com
4. Ultrasonic Sensor (HC-SR04) Datasheet
5. Research papers on obstacle detection and assistive devices
6. Online tutorials and project guides from websites like Electronics Hub, Circuit Digest, and Random Nerd Tutorials.